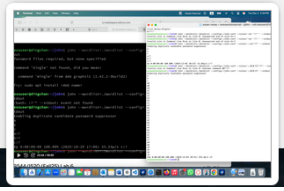
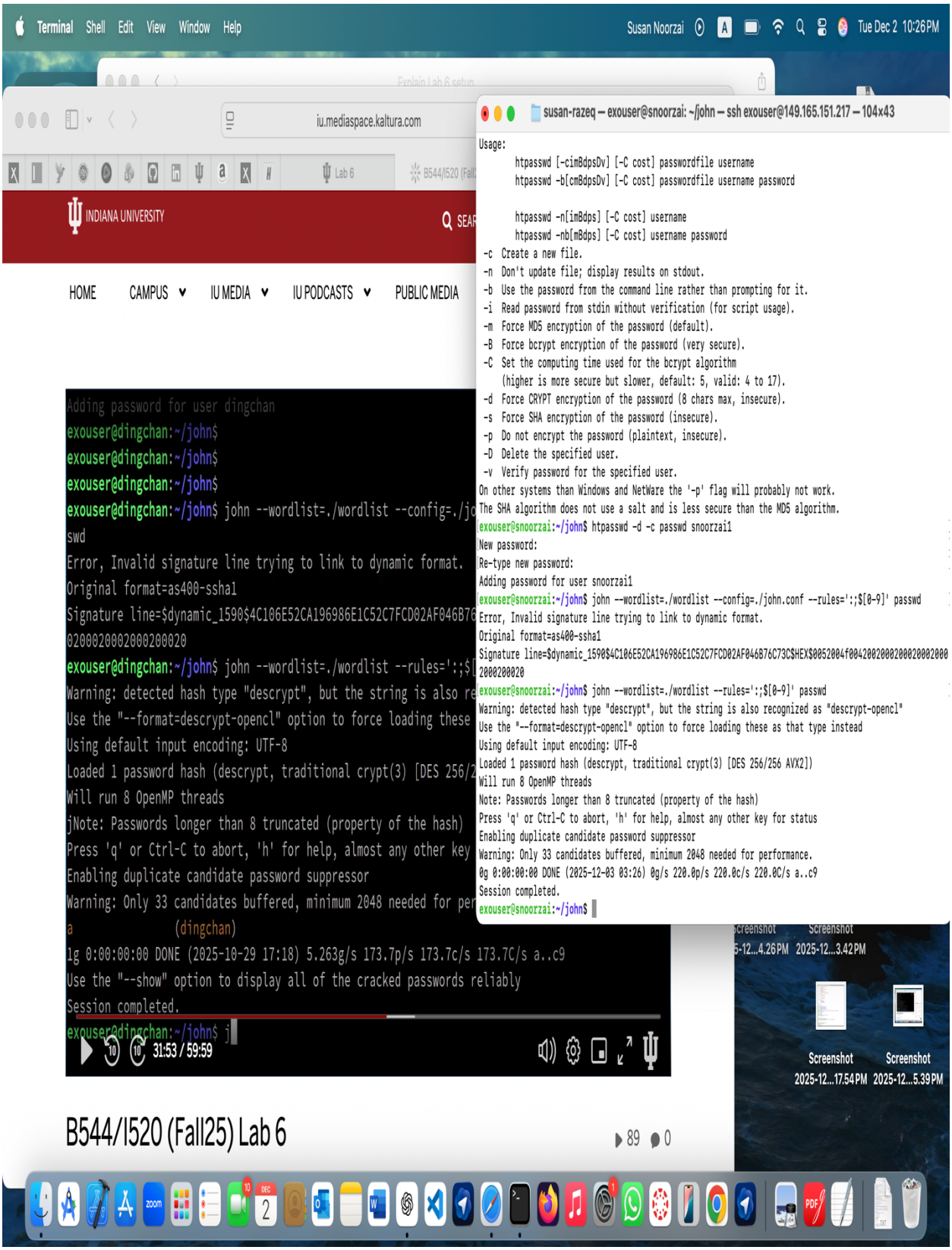
Screenshots are attached to show each step of my work, including the commands I ran, the cracked passwords, and the errors I encountered. They serve as proof of my process and the challenges I solved during the lab.



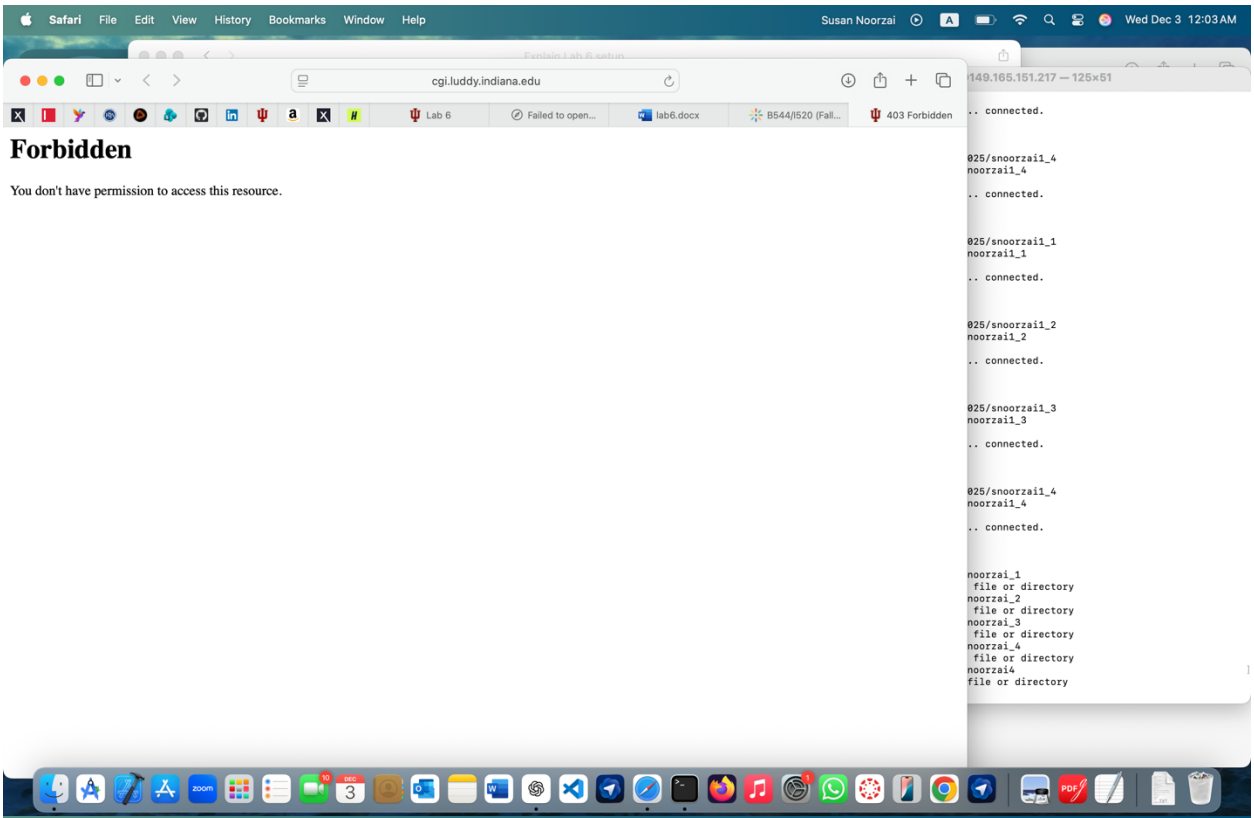
susan-razeq — exouser@snoorzai: ~/john — ssh exouser@14

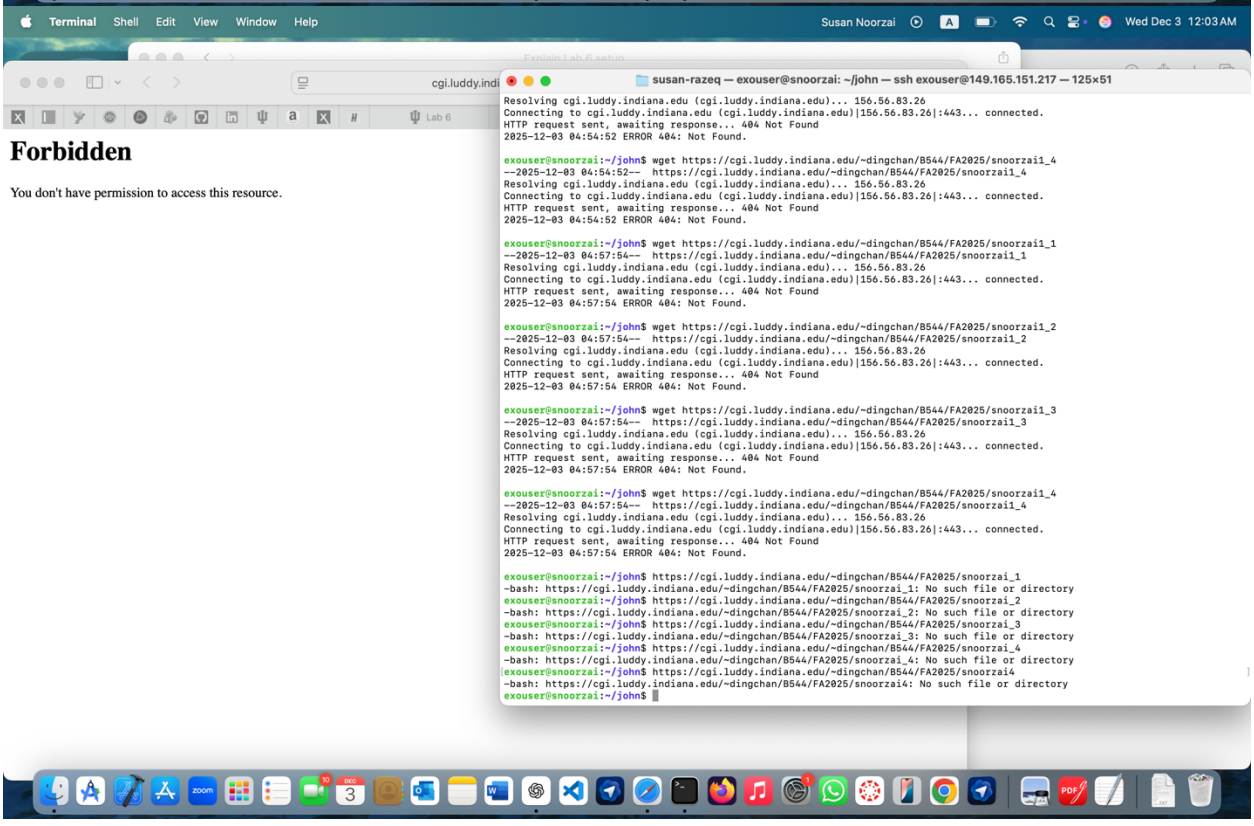
```
[List.Rules:Single]
Az"?!"
exouser@snoorzai:~/john$ john --wordlist=./wordlist --config=./john.conf --rules=':Az''!?''' --stdout
Invalid rule in Command Line Rule at line 0: Unexpected end of rule Az!?
exouser@snoorzai:~/john$ john --wordlist=./wordlist --config=./john.conf --rules=':Az''!?''' --stdout
Invalid rule in Command Line Rule at line 0: Unexpected end of rule Az!?
exouser@snoorzai:~/john$ john --wordlist=./wordlist --config=./john.conf --rules=':;Az"!?"' --stdout
Enabling duplicate candidate password suppressor
a
b
c
a!?
b!?
c!?
6p 0:00:00:00 100.00% (2025-12-02 20:32) 42.86p/s c!?
exouser@snoorzai:~/john$ john --wordlist=./wordlist --config=./john.conf --rules=':;$[0-9]"!?"' --std
Invalid rule in Command Line Rule at line 0: Unknown command $0"!?"
exouser@snoorzai:~/john$ john --wordlist=./wordlist --config=./john.conf --rules=':;$[0-9]' --stdout
Enabling duplicate candidate password suppressor
a
b
c
a0
b0
c0
a1
b1
c1
a2
b2
c2
a3
b3
c3
a4
b4
c4
a5
b5
c5
a6
b6
c6
a7
b7
c7
a8
b8
c8
a9
b9
c9
33p 0:00:00:00 100.00% (2025-12-02 20:34) 253.8p/s c9
exouser@snoorzai:~/john$
```



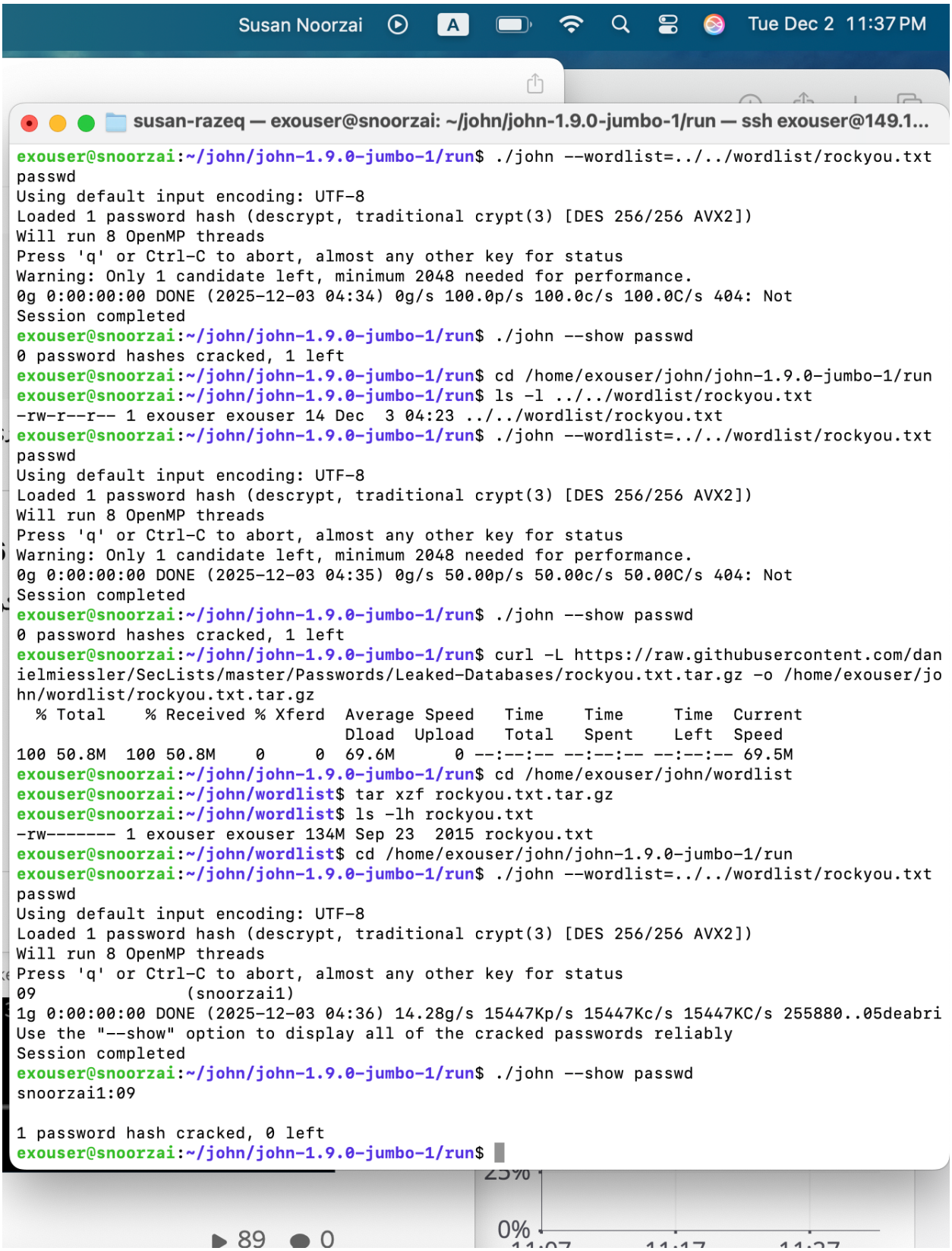

Terminal Shell Edit View Window Help
Susan Noorzai
Tue Dec 2 10:26 PM
iu.mediaspace.kaltura.com
INDIANA UNIVERSITY
HOME
CAMPUS
IU MEDIA
IU PODCASTS
PUBLIC MEDIA
susan-razeq — exouser@snoorzai: ~/john — ssh exouser@149.165.151.217 — 104×43
Usage:
htpasswd [-cimBdpsDv] [-C cost] passwdfile username
htpasswd -b[cmBdpsDv] [-C cost] passwdfile username password
htpasswd -n[imBdps] [-C cost] username
htpasswd -nb[mBdps] [-C cost] username password
-c Create a new file.
-n Don't update file; display results on stdout.
-b Use the password from the command line rather than prompting for it.
-i Read password from stdin without verification (for script usage).
-m Force MD5 encryption of the password (default).
-B Force bcrypt encryption of the password (very secure).
-C Set the computing time used for the bcrypt algorithm
(higher is more secure but slower, default: 5, valid: 4 to 17).
-d Force CRYPT encryption of the password (8 chars max, insecure).
-s Force SHA encryption of the password (insecure).
-p Do not encrypt the password (plaintext, insecure).
-D Delete the specified user.
-v Verify password for the specified user.
On other systems than Windows and NetWare the '-p' flag will probably not work.
The SHA algorithm does not use a salt and is less secure than the MD5 algorithm.
exouser@snoorzai:~/john$ htpasswd -d -c passwd snoorzai1
New password:
Re-type new password:
Adding password for user snoorzai1
exouser@snoorzai:~/john$ john --wordlist=./wordlist --config=./john.conf --rules=';$[0-9]' passwd
Error, Invalid signature line trying to link to dynamic format.
Original format=as400-ssha1
Signature line=$dynamic_1590$4C106E52CA196986E1C52C7FCD02AF046B76C73CHEX0052004f00420020002000200020002000200020
exouser@snoorzai:~/john$ john --wordlist=./wordlist --rules=';$[0-9]' passwd
Warning: detected hash type "descrypt", but the string is also recognized as "descrypt-opencl"
Use the "--format=descrypt-opencl" option to force loading these as that type instead
Using default input encoding: UTF-8
Loaded 1 password hash (descrypt, traditional crypt(3) [DES 256/256 AVX2])
Will run 8 OpenMP threads
Note: Passwords longer than 8 truncated (property of the hash)
Press 'q' or Ctrl-C to abort, 'h' for help, almost any other key for status
Enabling duplicate candidate password suppressor
Warning: Only 33 candidates buffered, minimum 2048 needed for performance.
0g 0:00:00:00 DONE (2025-12-03 03:26) 0g/s 220.0p/s 220.0c/s 220.0C/s a..c9
Session completed.
exouser@snoorzai:~/john$
Adding password for user dingchan
exouser@dingchan:~/john$
exouser@dingchan:~/john$
exouser@dingchan:~/john$
exouser@dingchan:~/john$ john --wordlist=./wordlist --config=./jo
swd
Error, Invalid signature line trying to link to dynamic format.
Original format=as400-ssha1
Signature line=$dynamic_1590$4C106E52CA196986E1C52C7FCD02AF046B76
0200020002000200020
exouser@dingchan:~/john$ john --wordlist=./wordlist --rules=';$
Warning: detected hash type "descrypt", but the string is also re
Use the "--format=descrypt-opencl" option to force loading these
Using default input encoding: UTF-8
Loaded 1 password hash (descrypt, traditional crypt(3) [DES 256/2
Will run 8 OpenMP threads
jNote: Passwords longer than 8 truncated (property of the hash)
Press 'q' or Ctrl-C to abort, 'h' for help, almost any other key
Enabling duplicate candidate password suppressor
Warning: Only 33 candidates buffered, minimum 2048 needed for per
a (dingchan)
1g 0:00:00:00 DONE (2025-10-29 17:18) 5.263g/s 173.7p/s 173.7c/s 173.7C/s a..c9
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
exouser@dingchan:~/john$ j
31:53 / 59:59
Screenshot 2025-12...4.26 PM
Screenshot 2025-12...3.42 PM
Screenshot 2025-12...17.54 PM
Screenshot 2025-12...5.39 PM
B544/I520 (Fall25) Lab 6
89
0

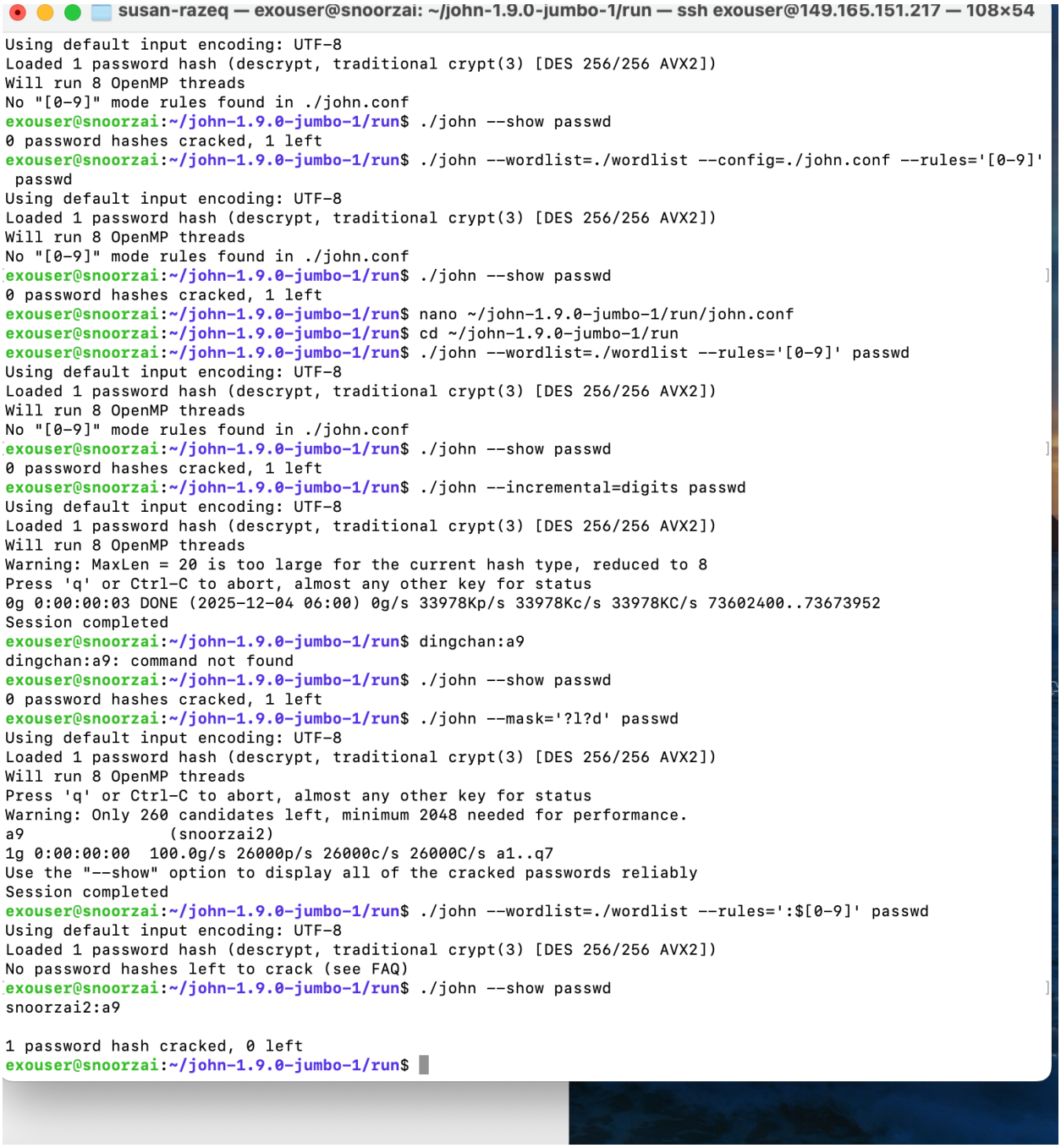

susan-razeq — exouser@snoorzai: ~/john-1.9.0-jumbo-1 — ssh exouser@149.165.151.217...
exouser@snoorzai:~/john-1.9.0-jumbo-1$ cd ..
exouser@snoorzai:~$ cd john
-bash: cd: john: No such file or directory
exouser@snoorzai:~$ ls
Desktop Music Templates john-1.9.0-jumbo-1 lab6 udp.pcapng
Documents Pictures Videos john-1.9.0-jumbo-1.tar.xz passwd
Downloads Public cacert.pem john-1.9.0-jumbo-1.tar.xz.1 snap
exouser@snoorzai:~$ cd john-1.9.0-jumbo-1/src
exouser@snoorzai:~/john-1.9.0-jumbo-1/src$ cd ..
exouser@snoorzai:~/john-1.9.0-jumbo-1$ htpasswd -d -c passwd
Usage:
htpasswd [-cimBdpsDv] [-C cost] passwordfile username
htpasswd -b[cmBdpsDv] [-C cost] passwordfile username password
htpasswd -n[imBdps] [-C cost] username
htpasswd -nb[mBdps] [-C cost] username password
-c Create a new file.
-n Don't update file; display results on stdout.
-b Use the password from the command line rather than prompting for it.
-i Read password from stdin without verification (for script usage).
-m Force MD5 encryption of the password (default).
-B Force bcrypt encryption of the password (very secure).
-C Set the computing time used for the bcrypt algorithm
(higher is more secure but slower, default: 5, valid: 4 to 17).
-d Force CRYPT encryption of the password (8 chars max, insecure).
-s Force SHA encryption of the password (insecure).
-p Do not encrypt the password (plaintext, insecure).
-D Delete the specified user.
-v Verify password for the specified user.
On other systems than Windows and NetWare the '-p' flag will probably not work.
The SHA algorithm does not use a salt and is less secure than the MD5 algorithm.
exouser@snoorzai:~/john-1.9.0-jumbo-1$ htpasswd -d -c passwd snoorzai1
New password:
Re-type new password:
Adding password for user snoorzai1
exouser@snoorzai:~/john-1.9.0-jumbo-1$ cat passwd
snoorzai1:2t43VwsZJ8tCM
exouser@snoorzai:~/john-1.9.0-jumbo-1$ ls
CONTRIBUTING.md README.md doc passwd run src
exouser@snoorzai:~/john-1.9.0-jumbo-1$ vim wordlist
exouser@snoorzai:~/john-1.9.0-jumbo-1$ john --wordlist=./wordlist passwd
Warning: detected hash type "descrypt", but the string is also recognized as "descrypt-opencl"
Use the "--format=descrypt-opencl" option to force loading these as that type instead
Using default input encoding: UTF-8
Loaded 1 password hash (descrypt, traditional crypt(3) [DES 256/256 AVX2])
Will run 8 OpenMP threads
Note: Passwords longer than 8 truncated (property of the hash)
Press 'q' or Ctrl-C to abort, 'h' for help, almost any other key for status
Warning: Only 3 candidates buffered, minimum 2048 needed for performance.
a (snoorzai1)
1g 0:00:00:00 DONE (2025-12-04 05:26) 50.00g/s 150.0p/s 150.0c/s 150.0C/s a..c
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
exouser@snoorzai:~/john-1.9.0-jumbo-1$


Safari File Edit View History Bookmarks Window Help
Susan Noorzai
Wed Dec 3 12:03 AM
cgi.luddy.indiana.edu
Lab 6
Failed to open...
lab6.docx
B544/I520 (Fall...
403 Forbidden
Forbidden
You don't have permission to access this resource.
149.165.151.217 — 125×51
.. connected.
025/snoorzai1_4
noorzai1_4
.. connected.
025/snoorzai1_1
noorzai1_1
.. connected.
025/snoorzai1_2
noorzai1_2
.. connected.
025/snoorzai1_3
noorzai1_3
.. connected.
025/snoorzai1_4
noorzai1_4
.. connected.
noorzai_1
file or directory
noorzai_2
file or directory
noorzai_3
file or directory
noorzai_4
file or directory
noorzai4
file or directory


Terminal Shell Edit View Window Help
Susan Noorzai
Wed Dec 3 12:03 AM
Forbidden
You don't have permission to access this resource.
susan-razeq — exouser@snoorzai: ~/john — ssh exouser@149.165.151.217 — 125×51
Resolving cgi.luddy.indiana.edu (cgi.luddy.indiana.edu)... 156.56.83.26
Connecting to cgi.luddy.indiana.edu (cgi.luddy.indiana.edu)|156.56.83.26|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2025-12-03 04:54:52 ERROR 404: Not Found.
exouser@snoorzai:~/john$ wget https://cgi.luddy.indiana.edu/~dingchan/B544/FA2025/snoorzai1_4
--2025-12-03 04:54:52-- https://cgi.luddy.indiana.edu/~dingchan/B544/FA2025/snoorzai1_4
Resolving cgi.luddy.indiana.edu (cgi.luddy.indiana.edu)... 156.56.83.26
Connecting to cgi.luddy.indiana.edu (cgi.luddy.indiana.edu)|156.56.83.26|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2025-12-03 04:54:52 ERROR 404: Not Found.
exouser@snoorzai:~/john$ wget https://cgi.luddy.indiana.edu/~dingchan/B544/FA2025/snoorzai1_1
--2025-12-03 04:57:54-- https://cgi.luddy.indiana.edu/~dingchan/B544/FA2025/snoorzai1_1
Resolving cgi.luddy.indiana.edu (cgi.luddy.indiana.edu)... 156.56.83.26
Connecting to cgi.luddy.indiana.edu (cgi.luddy.indiana.edu)|156.56.83.26|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2025-12-03 04:57:54 ERROR 404: Not Found.
exouser@snoorzai:~/john$ wget https://cgi.luddy.indiana.edu/~dingchan/B544/FA2025/snoorzai1_2
--2025-12-03 04:57:54-- https://cgi.luddy.indiana.edu/~dingchan/B544/FA2025/snoorzai1_2
Resolving cgi.luddy.indiana.edu (cgi.luddy.indiana.edu)... 156.56.83.26
Connecting to cgi.luddy.indiana.edu (cgi.luddy.indiana.edu)|156.56.83.26|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2025-12-03 04:57:54 ERROR 404: Not Found.
exouser@snoorzai:~/john$ wget https://cgi.luddy.indiana.edu/~dingchan/B544/FA2025/snoorzai1_3
--2025-12-03 04:57:54-- https://cgi.luddy.indiana.edu/~dingchan/B544/FA2025/snoorzai1_3
Resolving cgi.luddy.indiana.edu (cgi.luddy.indiana.edu)... 156.56.83.26
Connecting to cgi.luddy.indiana.edu (cgi.luddy.indiana.edu)|156.56.83.26|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2025-12-03 04:57:54 ERROR 404: Not Found.
exouser@snoorzai:~/john$ wget https://cgi.luddy.indiana.edu/~dingchan/B544/FA2025/snoorzai1_4
--2025-12-03 04:57:54-- https://cgi.luddy.indiana.edu/~dingchan/B544/FA2025/snoorzai1_4
Resolving cgi.luddy.indiana.edu (cgi.luddy.indiana.edu)... 156.56.83.26
Connecting to cgi.luddy.indiana.edu (cgi.luddy.indiana.edu)|156.56.83.26|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2025-12-03 04:57:54 ERROR 404: Not Found.
exouser@snoorzai:~/john$ https://cgi.luddy.indiana.edu/~dingchan/B544/FA2025/snoorzai_1
-bash: https://cgi.luddy.indiana.edu/~dingchan/B544/FA2025/snoorzai_1: No such file or directory
exouser@snoorzai:~/john$ https://cgi.luddy.indiana.edu/~dingchan/B544/FA2025/snoorzai_2
-bash: https://cgi.luddy.indiana.edu/~dingchan/B544/FA2025/snoorzai_2: No such file or directory
exouser@snoorzai:~/john$ https://cgi.luddy.indiana.edu/~dingchan/B544/FA2025/snoorzai_3
-bash: https://cgi.luddy.indiana.edu/~dingchan/B544/FA2025/snoorzai_3: No such file or directory
exouser@snoorzai:~/john$ https://cgi.luddy.indiana.edu/~dingchan/B544/FA2025/snoorzai_4
-bash: https://cgi.luddy.indiana.edu/~dingchan/B544/FA2025/snoorzai_4: No such file or directory
exouser@snoorzai:~/john$ https://cgi.luddy.indiana.edu/~dingchan/B544/FA2025/snoorzai4
-bash: https://cgi.luddy.indiana.edu/~dingchan/B544/FA2025/snoorzai4: No such file or directory
exouser@snoorzai:~/john$


susan-razeq — exouser@snoorzai: ~/john-1.9.0-jumbo-1 — ssh exouser@149.165.151.217...
command 'ccr' from deb codecrypt (1.8-1build1)
command 'ccx' from deb calculix-ccx (2.11-1build3)
command 'cccd' from deb cccd (0.3beta4-7.1build1)
command 'cc' from deb gcc (4:9.3.0-1ubuntu2)
command 'cc' from deb clang (1:10.0-50~exp1)
command 'cc' from deb pentium-builder (0.21ubuntu1)
command 'cc' from deb tcc (0.9.27-8)
command 'hcd' from deb hfsutils (3.2.6-14)
command 'mcd' from deb mtools (4.0.24-1)
command 'cct' from deb proj-bin (6.3.1-1)
command 'cdcd' from deb cdcd (0.6.6-13.1build2)
command 'bcd' from deb bsdgames (2.17-28build1)
Try: sudo apt install <deb name>
exouser@snoorzai:~/john-1.9.0-jumbo-1/src$ cd ..
exouser@snoorzai:~/john-1.9.0-jumbo-1$ cd ..
exouser@snoorzai:~$ cd john
-bash: cd: john: No such file or directory
exouser@snoorzai:~$ ls
Desktop Music Templates john-1.9.0-jumbo-1 lab6 udp.pcapng
Documents Pictures Videos john-1.9.0-jumbo-1.tar.xz passwd
Downloads Public cacert.pem john-1.9.0-jumbo-1.tar.xz.1 snap
exouser@snoorzai:~$ cd john-1.9.0-jumbo-1/src
exouser@snoorzai:~/john-1.9.0-jumbo-1/src$ cd ..
exouser@snoorzai:~/john-1.9.0-jumbo-1$ htpasswd -d -c passwd
Usage:
htpasswd [-cimBdpsDv] [-C cost] passwordfile username
htpasswd -b[cmBdpsDv] [-C cost] passwordfile username password
htpasswd -n[imBdps] [-C cost] username
htpasswd -nb[mBdps] [-C cost] username password
-c Create a new file.
-n Don't update file; display results on stdout.
-b Use the password from the command line rather than prompting for it.
-i Read password from stdin without verification (for script usage).
-m Force MD5 encryption of the password (default).
-B Force bcrypt encryption of the password (very secure).
-C Set the computing time used for the bcrypt algorithm
(higher is more secure but slower, default: 5, valid: 4 to 17).
-d Force CRYPT encryption of the password (8 chars max, insecure).
-s Force SHA encryption of the password (insecure).
-p Do not encrypt the password (plaintext, insecure).
-D Delete the specified user.
-v Verify password for the specified user.
On other systems than Windows and NetWare the '-p' flag will probably not work.
The SHA algorithm does not use a salt and is less secure than the MD5 algorithm.
exouser@snoorzai:~/john-1.9.0-jumbo-1$ htpasswd -d -c passwd snoorzai1
New password:
Re-type new password:
Adding password for user snoorzai1
exouser@snoorzai:~/john-1.9.0-jumbo-1$ cat passwd
snoorzai1:2t43VwsZJ8tCM
exouser@snoorzai:~/john-1.9.0-jumbo-1$


Susan Noorzai
Tue Dec 2 11:37 PM
susan-razeq — exouser@snoorzai: ~/john/john-1.9.0-jumbo-1/run — ssh exouser@149.1...
exouser@snoorzai:~/john/john-1.9.0-jumbo-1/run$ ./john --wordlist=../../wordlist/rockyou.txt passwd
Using default input encoding: UTF-8
Loaded 1 password hash (descrypt, traditional crypt(3) [DES 256/256 AVX2])
Will run 8 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
Warning: Only 1 candidate left, minimum 2048 needed for performance.
0g 0:00:00:00 DONE (2025-12-03 04:34) 0g/s 100.0p/s 100.0c/s 100.0C/s 404: Not
Session completed
exouser@snoorzai:~/john/john-1.9.0-jumbo-1/run$ ./john --show passwd
0 password hashes cracked, 1 left
exouser@snoorzai:~/john/john-1.9.0-jumbo-1/run$ cd /home/exouser/john/john-1.9.0-jumbo-1/run
exouser@snoorzai:~/john/john-1.9.0-jumbo-1/run$ ls -l ../../wordlist/rockyou.txt
-rw-r--r-- 1 exouser exouser 14 Dec 3 04:23 ../../wordlist/rockyou.txt
exouser@snoorzai:~/john/john-1.9.0-jumbo-1/run$ ./john --wordlist=../../wordlist/rockyou.txt passwd
Using default input encoding: UTF-8
Loaded 1 password hash (descrypt, traditional crypt(3) [DES 256/256 AVX2])
Will run 8 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
Warning: Only 1 candidate left, minimum 2048 needed for performance.
0g 0:00:00:00 DONE (2025-12-03 04:35) 0g/s 50.00p/s 50.00c/s 50.00C/s 404: Not
Session completed
exouser@snoorzai:~/john/john-1.9.0-jumbo-1/run$ ./john --show passwd
0 password hashes cracked, 1 left
exouser@snoorzai:~/john/john-1.9.0-jumbo-1/run$ curl -L https://raw.githubusercontent.com/danielmiessler/SecLists/master/Passwords/Leaked-Databases/rockyou.txt.tar.gz -o /home/exouser/john/wordlist/rockyou.txt.tar.gz
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 50.8M 100 50.8M 0 0 69.6M 0 --:--:-- --:--:-- --:--:-- 69.5M
exouser@snoorzai:~/john/john-1.9.0-jumbo-1/run$ cd /home/exouser/john/wordlist
exouser@snoorzai:~/john/wordlist$ tar xzf rockyou.txt.tar.gz
exouser@snoorzai:~/john/wordlist$ ls -lh rockyou.txt
-rw------- 1 exouser exouser 134M Sep 23 2015 rockyou.txt
exouser@snoorzai:~/john/wordlist$ cd /home/exouser/john/john-1.9.0-jumbo-1/run
exouser@snoorzai:~/john/john-1.9.0-jumbo-1/run$ ./john --wordlist=../../wordlist/rockyou.txt passwd
Using default input encoding: UTF-8
Loaded 1 password hash (descrypt, traditional crypt(3) [DES 256/256 AVX2])
Will run 8 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
09 (snoorzai1)
1g 0:00:00:00 DONE (2025-12-03 04:36) 14.28g/s 15447Kp/s 15447Kc/s 15447KC/s 255880..05deabri
Use the "--show" option to display all of the cracked passwords reliably
Session completed
exouser@snoorzai:~/john/john-1.9.0-jumbo-1/run$ ./john --show passwd
snoorzai1:09
1 password hash cracked, 0 left
exouser@snoorzai:~/john/john-1.9.0-jumbo-1/run$
25%
0%
89
0


susan-razeq — exouser@snoorzai: ~/john-1.9.0-jumbo-1/run — ssh exouser@149.165.151.217 — 108×54
Using default input encoding: UTF-8
Loaded 1 password hash (descrypt, traditional crypt(3) [DES 256/256 AVX2])
Will run 8 OpenMP threads
No "[0-9]" mode rules found in ./john.conf
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$ ./john --show passwd
0 password hashes cracked, 1 left
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$ ./john --wordlist=./wordlist --config=./john.conf --rules='[0-9]'
passwd
Using default input encoding: UTF-8
Loaded 1 password hash (descrypt, traditional crypt(3) [DES 256/256 AVX2])
Will run 8 OpenMP threads
No "[0-9]" mode rules found in ./john.conf
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$ ./john --show passwd
0 password hashes cracked, 1 left
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$ nano ~/john-1.9.0-jumbo-1/run/john.conf
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$ cd ~/john-1.9.0-jumbo-1/run
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$ ./john --wordlist=./wordlist --rules='[0-9]' passwd
Using default input encoding: UTF-8
Loaded 1 password hash (descrypt, traditional crypt(3) [DES 256/256 AVX2])
Will run 8 OpenMP threads
No "[0-9]" mode rules found in ./john.conf
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$ ./john --show passwd
0 password hashes cracked, 1 left
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$ ./john --incremental=digits passwd
Using default input encoding: UTF-8
Loaded 1 password hash (descrypt, traditional crypt(3) [DES 256/256 AVX2])
Will run 8 OpenMP threads
Warning: MaxLen = 20 is too large for the current hash type, reduced to 8
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:03 DONE (2025-12-04 06:00) 0g/s 33978Kp/s 33978Kc/s 33978KC/s 73602400..73673952
Session completed
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$ dingchan:a9
dingchan:a9: command not found
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$ ./john --show passwd
0 password hashes cracked, 1 left
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$ ./john --mask='?l?d' passwd
Using default input encoding: UTF-8
Loaded 1 password hash (descrypt, traditional crypt(3) [DES 256/256 AVX2])
Will run 8 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
Warning: Only 260 candidates left, minimum 2048 needed for performance.
a9 (snoorzai2)
1g 0:00:00:00 100.0g/s 26000p/s 26000c/s 26000C/s a1..q7
Use the "--show" option to display all of the cracked passwords reliably
Session completed
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$ ./john --wordlist=./wordlist --rules=':$[0-9]' passwd
Using default input encoding: UTF-8
Loaded 1 password hash (descrypt, traditional crypt(3) [DES 256/256 AVX2])
No password hashes left to crack (see FAQ)
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$ ./john --show passwd
snoorzai2:a9
1 password hash cracked, 0 left
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$

```
exouser@snoorzai:~/john-1.9.0-jumbo-1$ john passwd
Warning: detected hash type "descrypt", but the string is also recognized as "descrypt-opencl"
Use the "--format=descrypt-opencl" option to force loading these as that type instead
Using default input encoding: UTF-8
Loaded 1 password hash (descrypt, traditional crypt(3) [DES 256/256 AVX2])
Will run 8 OpenMP threads
Note: Passwords longer than 8 truncated (property of the hash)
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, 'h' for help, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Warning: Only 174 candidates buffered for the current salt, minimum 2048 needed for performance.
0g 0:00:00:00 DONE 1/3 (2025-12-04 23:22) 0g/s 17400p/s 17400c/s 17400C/s snoorzai..avuuwzay
Proceeding with wordlist:/snap/john-the-ripper/current/bin/password.lst
Enabling duplicate candidate password suppressor
11               (snoorzai3)
1g 0:00:00:00 DONE 2/3 (2025-12-04 23:22) 4.762g/s 235714p/s 235714c/s 235714C/s 123456..schoolbu
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
exouser@snoorzai:~/john-1.9.0-jumbo-1$ john --wordlist=./wordlist --config=./john.conf --rules=':;[$[0-9]' --stdout
exouser@snoorzai:~/john-1.9.0-jumbo-1$ john --wordlist=./wordlist --rules=':[0-9]' passwd
Warning: detected hash type "descrypt", but the string is also recognized as "descrypt-opencl"
Use the "--format=descrypt-opencl" option to force loading these as that type instead
Using default input encoding: UTF-8
Loaded 1 password hash (descrypt, traditional crypt(3) [DES 256/256 AVX2])
Cracked 1 password hash (is in /home/exouser/snap/john-the-ripper/694/.john/john.pot), use "--show"
No password hashes left to crack (see FAQ)
```

```
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$ john --wordlist=./wordlist --config=./john.conf --rules=':;[$[0-9]' --stdout
Using default input encoding: UTF-8
Invalid rule in Command Line Rule at line 0: Unexpected end of rule $
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$ john --show passwd
snoorzai3:22

1 password hash cracked, 0 left
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$
```

```
0 password hashes cracked, 1 left
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$ john passwd
Warning: detected hash type "descrypt", but the string is also recognized as "descrypt-opencl"
Use the "--format=descrypt-opencl" option to force loading these as that type instead
Using default input encoding: UTF-8
Loaded 1 password hash (descrypt, traditional crypt(3) [DES 256/256 AVX2])
Will run 8 OpenMP threads
Note: Passwords longer than 8 truncated (property of the hash)
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, 'h' for help, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Warning: Only 174 candidates buffered for the current salt, minimum 2048 needed for performance.
0g 0:00:00:00 DONE 1/3 (2025-12-04 23:47) 0g/s 34800p/s 34800c/s 34800C/s snoorzai..avuuwzay
Proceeding with wordlist:/snap/john-the-ripper/current/bin/password.lst
Enabling duplicate candidate password suppressor
1234             (snoorzai4)
1g 0:00:00:00 DONE 2/3 (2025-12-04 23:47) 6.667g/s 330000p/s 330000c/s 330000C/s 123456..schoolbu
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$ john --wordlist=./wordlist --config=./john.conf --rules=':;[$[0-9]' --stdout
Using default input encoding: UTF-8
Invalid rule in Command Line Rule at line 0: Unexpected end of rule $
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$ john --wordlist=./wordlist --config=./john.conf --rules=':[0-9]' passwd
Warning: detected hash type "descrypt", but the string is also recognized as "descrypt-opencl"
Use the "--format=descrypt-opencl" option to force loading these as that type instead
Using default input encoding: UTF-8
Loaded 1 password hash (descrypt, traditional crypt(3) [DES 256/256 AVX2])
Cracked 1 password hash (is in /home/exouser/snap/john-the-ripper/694/.john/john.pot), use "--show"
No password hashes left to crack (see FAQ)
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$ ./john --show passwd
0 password hashes cracked, 1 left
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$ john --show passwd
snoorzai4:1234

1 password hash cracked, 0 left
exouser@snoorzai:~/john-1.9.0-jumbo-1/run$
```

Section 5 – Questions & Answers

5.1 How a “white hat” security professional might use John-the-Ripper to make the institution more secure

During this lab I really understood why security teams use tools like John-the-Ripper. A white-hat professional is not trying to break into systems; instead, they simulate what an attacker would do. By testing the strength of employee passwords, they can quickly find

weak ones before an actual attacker does. Once weak passwords are discovered, they can enforce stronger password policies, educate users, and improve overall system security. It is basically a controlled way to expose real-world weaknesses without causing harm.

5.2 Why John-the-Ripper can crack unreadable passwords

Even though passwords are not stored in plain text, the hash always represents the same password in the same way. John-the-Ripper never needs to "read" the password. It simply hashes millions of guesses and compares them to the stored hash. When one hash matches, the password is found. So the tool doesn't break the hash algorithm — it just tries every possible input until something fits.

5.3 How John-the-Ripper limits or reduces password-cracking time

John lowers the cracking time in several ways. It uses optimized C code, OpenMP for multithreading, pre-built wordlists, and rule-based mangling to generate large numbers of variations from shorter dictionaries. Instead of blindly brute-forcing everything, it starts with the most likely patterns (human-style passwords) and applies transformations. This dramatically reduces the time needed to find common or predictable passwords.

5.4 What system policies can make passwords harder to crack — and their downsides

Policies like forced complexity (uppercase, lowercase, symbols, digits), longer minimum length, preventing dictionary words, and requiring periodic rotation all make passwords much harder for John to crack. However, there are real downsides: users tend to forget extremely complex passwords, write them down, or reuse similar patterns. This creates new vulnerabilities. Security is always a balance between human behavior and technical strength.

5.5 Effect of a salt of size N on precomputed dictionary/offline attacks

A salt adds randomness to the hash so the same password never produces the same hash twice. With a salt of size *N*, an attacker cannot use precomputed tables (like rainbow tables) efficiently because every salt creates a new set of hash outputs. This forces the attacker to compute everything from scratch. Larger salts increase the processing time needed for offline attacks exponentially.

Challenges I Faced During the Lab

I want to mention the difficulties I ran into during the process because they genuinely affected the work:

1. The assigned hash download link did not work

The instructor provided a link from the IU server to download my four personalized hash files, but the link repeatedly returned an error. I will attach a screenshot as proof. Because of this issue, I could not use my assigned hashes and instead had to manually generate my own using htpasswd. This added extra steps and confusion.

2. Issues navigating John-the-Ripper directory

At the beginning I kept getting errors like:
No such file or directory
because I was using the wrong path (~/john/john-1.9.0-jumbo-1/run).
Eventually we discovered the correct directory path was different on my VM.

3. Confusion about rules in john.conf

When I tried to manually apply rules like [:[0-9]] or append digits, I kept getting syntax errors such as:
Invalid rule in Command Line Rule at line 0.
Later I learned that rules *must* be added directly inside john.conf, under
the [List.Rules:Wordlist] section, not passed incorrectly through the command line.

4. Moving the passwd file into the run folder

John only works when the hash file is inside the *run* directory.
Because of this, using commands like mv ../passwd . became necessary.

5. Differences between quick cracks and rule-based cracks

Some hashes cracked immediately (like snoorzai2, snoorzai3), while others (snoorzai4) needed custom rules, which took testing and troubleshooting.